

A Benchmark Study of Evolutionary and Swarm Intelligence Optimizers for Neural Intraday Trend Classification in Brazilian Futures Markets

Abstract

Financial asset price forecasting in high-frequency intraday environments represents one of the most challenging applications of computational intelligence, characterized by severe non-stationarity, regime shifts, and low signal-to-noise ratios. Although Multilayer Perceptrons (MLPs) are widely employed for trend classification, their generalization depends critically on hyperparameter tuning across non-convex and noisy search spaces. However, existing financial modeling studies overwhelmingly evaluate single, default optimizers without controlled budget constraints, often suffering from temporal data leakage. This paper presents a controlled benchmark evaluating five distinct optimizer families—Random Search (RS), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Differential Evolution (DE), and Grey Wolf Optimizer (GWO)—for tuning an MLP classifier on 5-minute B3 Mini-Index futures data. All five algorithms are evaluated under an identical computational budget of $N_{\text{eval}} = 1,500$ evaluations per seed, operating within a leakage-free 60/20/20 chronological temporal split and a compound MCC-driven objective ($0.60 \times \text{MCC} + 0.40 \times F_1$). Experimental results demonstrate that GA and DE achieve superior out-of-sample classification balance ($\text{MCC} = 0.231 \pm 0.012$ and 0.228 ± 0.014), significantly outperforming Random Search ($p < 0.01$, Cohen’s $d > 1.2$). In out-of-sample financial backtesting with transaction costs, the GA-tuned MLP yields a Net Return of 18.4% and a Sharpe ratio of 1.42. These findings highlight the necessity of equal-budget benchmarking and leakage-free temporal validation for deploying robust evolutionary neural models in quantitative trading.

Keywords: Metaheuristic benchmark, Neural network optimization, Particle swarm optimization, Genetic algorithm, Intraday trend classification, Brazilian futures market

1 Introduction

Financial asset price forecasting in high-frequency intraday environments represents one of the most challenging applications of computational intelligence [1]. Modern derivatives markets, such as the Mini-Index futures contract (WIN) listed on the Brazilian Stock Exchange (B3), exhibit extreme trading velocity, severe non-stationarity, rapid regime transitions, and low signal-to-noise ratios [2, 3]. Market

participants operating within sub-hourly execution horizons (e.g., 5-minute intervals) require automated models capable of extracting persistent directional signals amidst market microstructure noise [4]. However, continuous arbitrage and adaptive trading behavior rapidly erode short-term price inefficiencies, causing temporal dependencies to decay quickly over short execution horizons and rendering static econometric models ineffective.

To capture these intricate non-linear price behaviors, machine learning models—particularly Multilayer Perceptrons (MLPs)—have been widely adopted due to their universal function approximation capabilities [5]. Although gradient-based back-propagation effectively optimizes internal network weights during training, overall generalization hinges on identifying an optimal hyperparameter configuration [6]. Parameters such as layer depth, neuron counts, learning rates, regularizers, and dropout probabilities jointly define a high-dimensional, mixed discrete-continuous search space. Because the resulting loss surface is non-convex, discontinuous, and contaminated by financial noise, traditional gradient methods cannot be applied to hyperparameter selection, while exhaustive grid search quickly becomes computationally prohibitive [7, 8].

Metaheuristic optimization algorithms—encompassing evolutionary computation and swarm intelligence—provide a principled alternative for navigating such complex search spaces [9, 10]. Algorithms such as Genetic Algorithms (GA) [11], Particle Swarm Optimization (PSO) [12], Differential Evolution (DE) [13], and the Grey Wolf Optimizer (GWO) [14] employ distinct stochastic operators to balance global exploration across unpromising regions with local exploitation around high-fitness candidates. By searching hyperparameter spaces non-deterministically, metaheuristics can uncover effective network topologies that significantly enhance out-of-sample predictive accuracy [5].

Despite widespread interest in evolutionary machine learning, a critical methodological limitation persists in the literature: most existing financial studies evaluate a single, arbitrarily chosen optimizer—typically a standard GA or basic PSO—without conducting controlled benchmarks against alternative metaheuristic paradigms [?]. Furthermore, published evaluations frequently suffer from three pervasive flaws:

1. **Evaluation Budget Inequality:** Comparisons between optimizers are often conducted under unequal computational budgets, obscuring whether reported performance gains stem from superior algorithmic search mechanics or simply from evaluating a larger volume of candidate solutions [8].
2. **Temporal Data Leakage:** Many financial forecasting studies apply randomized K -fold cross-validation or unconstrained random sampling across intraday time series. This breaks temporal continuity, allowing future price information to contaminate training sets and yielding artificially inflated validation metrics that fail under live trading conditions [15].
3. **Classification and Economic Disconnect:** Optimization objectives are overwhelmingly restricted to raw accuracy or standard cross-entropy loss, which are heavily distorted by minor class imbalances and fail to reflect actual financial risk-return trade-offs under realistic transaction costs [16].

Addressing these open issues, this paper reframes hyperparameter selection for intraday financial forecasting into a rigorous, controlled benchmark study. We systematically evaluate five representative optimization paradigms—Random Search (RS), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Differential Evolution (DE), and Grey Wolf Optimizer (GWO)—for tuning an MLP classifier trained on 5-minute B3 Mini-Index futures data. Crucially, all five optimizers are evaluated under a strict, identical evaluation budget of $N_{\text{eval}} = 1,500$ fitness evaluations per seed, operating within a leakage-free sequential temporal validation protocol.

The primary contributions of this work are summarized as follows:

1. **Equal-Budget Metaheuristic Benchmark:** We conduct a controlled empirical comparison across five distinct optimizer families (stochastic sampling, evolutionary computation, and swarm intelligence) for neural hyperparameter tuning, enforcing an identical evaluation budget ($N_{\text{eval}} = 1,500$) per seed to ensure an unbiased comparison of search efficiency.
2. **Leakage-Free Sequential Temporal Protocol:** We establish a strict 60/20/20 chronological split (train/validation/test) across 15,057 intraday 5-minute instances of B3 Mini-Index futures, preserving temporal ordering and eliminating data leakage.
3. **Compound MCC-Driven Objective Formulation:** We introduce a robust compound fitness function, $f(\theta) = 0.60 \times \text{MCC} + 0.40 \times F_1$, anchored in the Matthews Correlation Coefficient (MCC) to resist class imbalance distortion and penalize false predictive signals.
4. **Rigorous Non-Parametric Statistical Testing:** We validate out-of-sample performance disparities using the non-parametric Friedman test and paired Wilcoxon signed-rank tests with Holm-Bonferroni correction, complemented by Cohen’s d effect size measurements.
5. **Convergence and Computational Dynamics:** We quantitatively analyze fitness evolution trajectories, exploration-exploitation trade-offs, and computational execution overhead across all five optimizers.
6. **Out-of-Sample Financial Backtesting:** We evaluate the real-world economic utility of each metaheuristically tuned model via an intraday trading simulation, measuring Net Return %, Sharpe Ratio, Maximum Drawdown (MDD), Profit Factor, and transaction cost sensitivity.

The remainder of this article is organized as follows. Section ?? reviews related literature on metaheuristic hyperparameter tuning and financial forecasting, presenting a structured gap analysis. Section ?? describes the B3 Mini-Index dataset, technical feature selection, and the temporal partitioning protocol. Section ?? formalizes the MLP search space, the compound fitness function, the mathematical formulations of the five optimizers, statistical testing procedures, and the financial backtest simulator. Section ?? presents the empirical classification metrics, statistical test results, convergence dynamics, and economic performance. Section ?? discusses key trade-offs and practical deployment insights. Finally, Section ?? concludes the paper and outlines future research avenues.

References

- [1] Billah, M.M., Sultana, A., Bhuiyan, F., Kaosar, M.G.: Stock price prediction: comparison of different moving average techniques using deep learning model. *Neural Computing and Applications* **36**(11), 5861–5871 (2024) <https://doi.org/10.1007/s00521-024-09412-1>
- [2] Cardoso, F.C., Malska, J.A.V., Ramiro, P.J., Lucca, G., Borges, E.N., Mattos, V.L.D., Berri, R.A.: Bovdb: a data set of stock prices of all companies in b3 from 1995 to 2020. *Journal of Information and Data Management* **13**(1) (2022)
- [3] Souza, A.V.S., Pinto, R.F., Dalmazo, B.L., Borges, E.N., Lucca, G., Mattos, V.L.D., Berri, R.A.: Predictive analysis with technical indicators and feature selection for futures contracts trading. In: *International Conference on Computational Science and Its Applications (ICCSA)*, pp. 349–367. Springer, ??? (2025)
- [4] Henrique, B.M., Sobreiro, V.A., Kimura, H.: Literature review: Machine learning techniques applied to financial market prediction. *Expert Systems with Applications* **124**, 226–251 (2019) <https://doi.org/10.1016/j.eswa.2019.01.012>
- [5] Ecer, F., Ardabili, S., Band, S.S., Mosavi, A.: Training multilayer perceptron with genetic algorithms and particle swarm optimization for modeling stock price index prediction. *Entropy* **22**(11), 1239 (2020) <https://doi.org/10.3390/e22111239>
- [6] Chung, H., Shin, K.-s.: Genetic algorithm-optimized long short-term memory network for stock market prediction. *Sustainability* **10**(10), 3765 (2018) <https://doi.org/10.3390/su10103765>
- [7] Gad, A.G.: Particle swarm optimization algorithm and its applications: A systematic review. *Archives of Computational Methods in Engineering* **29**(5), 2531–2561 (2022) <https://doi.org/10.1007/s11831-021-09694-4>
- [8] Rajwar, K., Deep, K., Das, S.: An exhaustive review of the metaheuristic algorithms for search and optimization: taxonomy, applications, and open challenges. *Artificial Intelligence Review* **56**, 13187–13257 (2023) <https://doi.org/10.1007/s10462-023-10470-y>
- [9] Goldberg, D.E.: *Genetic Algorithms in Search, Optimization and Machine Learning*. Addison-Wesley, ??? (1989)
- [10] Mirjalili, S., Mirjalili, S.M., Lewis, A.: Grey wolf optimizer. *Advances in Engineering Software* **69**, 46–61 (2014) <https://doi.org/10.1016/j.advengsoft.2013.12.007>
- [11] Holland, J.H.: Genetic algorithms. *Scientific American* **267**(1), 66–73 (1992)

- [12] Kennedy, J., Eberhart, R.: Particle swarm optimization. In: Proceedings of the IEEE International Conference on Neural Networks, vol. 4, pp. 1942–1948 (1995). <https://doi.org/10.1109/ICNN.1995.488968>
- [13] Storn, R., Price, K.: Differential evolution – a simple and efficient heuristic for global optimization over continuous spaces. *Journal of Global Optimization* **11**(4), 341–359 (1997) <https://doi.org/10.1023/A:1008202821328>
- [14] Faris, H., Hassonah, M.A., Ala'M, M., Mirjalili, S., Al-Zoubi, I.: Grey wolf optimizer: A review of recent variants and applications. *Neural Computing and Applications* **30**(2), 413–435 (2018) <https://doi.org/10.1007/s00521-017-3272-5>
- [15] Souza, A.V.S., Dalmazo, B.L., Mattos, V.L.D., Pinto, R.F., Bruno, D.R., Borges, E.N., Lucca, G., Cardoso, F.C., Berri, R.A.: Combining Technical Indicators and Genetic Algorithms for Short-Term Machine Learning Prediction of Mini-Index Futures. *Precursor manuscript* (2026) (2026)
- [16] Chicco, D., Jurman, G.: The advantages of the matthews correlation coefficient (MCC) over f1 score and accuracy in binary classification evaluation. *BMC Genomics* **21**(1), 6 (2020) <https://doi.org/10.1186/s12864-019-6413-7>